

⚡ 05. Mongoose Queries & Update Comparisons

🔍 Querying in Mongoose (Thennables)

Mongoose queries native Promise nahi hoti, balki **Thennables (Query Objects)** hoti hain jinko `await` ya `.then()` se resolve kar sakte hain:

```
index.js - Query Methods

// 1. Saare matching documents array form me
const allUsers = await User.find({ age: { $gt: 20 } });
// 2. Single document ya null
const user = await User.findOne({ email: "amit@gmail.com" });
// 3. Primary key _id se dhoondhna
const userById = await User.findById("65c8a1f2 ... ");
```

⚖️ Updates: `updateOne` vs `findOneAndUpdate` (Critical Difference!)

Method	Return Value	Use Case
<code>User.updateOne()</code> / <code>updateMany()</code>	Metadata only: { acknowledged: true, modifiedCount: 1 }	Sirf modification status dekhna ho, document return nahi chahiye.
<code>User.findOneAndUpdate()</code> / <code>findByIdAndUpdate()</code>	Actual Document! (Default: old document; Pass { new: true } for updated doc)	API response ya UI me updated document render karna ho.

🗑️ Deletions: `deleteOne` vs `findByIdAndDelete`

Method	Return Value	Description
<code>User.deleteOne()</code>	{ acknowledged: true, deletedCount: 1 }	Sirf delete status return karta hai.
<code>User.findByIdAndDelete(id)</code>	Deleted Document Object	Deleted document ko return karta hai (undo/log ke liye useful).

Schema Validations & The runValidators Trap

Schema Validations in Depth

```
models/book.js — Advanced Schema Validations

const bookSchema = new mongoose.Schema({
  title: { type: String, required: [true, "Title is required!"], maxLength: 50 },
  price: { type: Number, min: [1, "Price must be at least ₹1"] },
  genre: { type: String, enum: ["fiction", "tech", "history"] },
  inStock: { type: Boolean, default: true }
});
```

 **THE runValidators TRAP (MOST COMMON INTERVIEW BUG):** By default Mongoose validation sirf .save() par chalti hai! Updates me validation chalane ke liye { runValidators: true } pass karna compulsory hai!

```
index.js — findByIdAndUpdate with runValidators

// Without runValidators: true, negative price (-500) bhi save ho jayega!
const updatedBook = await Book.findByIdAndUpdate(
  id,
  { price: -500 },
  { new: true, runValidators: true } // ➡ Triggers schema validation error!
);
```

Validation Summary Checklist:

- required: [true, "Custom message"] → Value hona mandatory hai.
- enum: ["val1", "val2"] → Sirf given options allow karta hai.
- min / max → Number aur Date types ke liye bounds set karta hai.